

Device power (grouped)

$$P_{\text{device}}(t) = P_{\text{base}} + P_{\text{screen}}(t) + P_{\text{comp}}(t) + P_{\text{radio}}(t) + P_{\text{user}}(t)$$

$$P_{\text{comp}}(t) = P_{\text{cpu}}(t) + P_{\text{gpu}}(t)$$

$$P_{\text{radio}}(t) = P_{\text{net}}(t) + P_{\text{gps}}(t)$$

$$P_{\text{user}}(t) = P_{\text{bg}}(t) + P_{\text{int}}(t)$$

Battery-side power

$$P_{\text{batt}}(t) = P_{\text{device}}(t) / \eta_{\text{PMIC}}$$

$$\eta_{\text{PMIC}} = \eta_0 \text{ (constant)}$$